

Exercise 1

The mtcars dataset comprises fuel consumption (mpg: Miles/(US) gallon) and 8 aspects of automobile design and performance for 32 automobiles. Specifically, the covariates are

- wt: Weight (1000 lbs)
- am: Transmission (0 = automatic, 1 = manual)
- cyl: Number of cylinders
- disp: Displacement (cu.in.)
- hp: Gross horsepower
- drat: Rear axle ratio
- qsec: 1/4 mile time
- vs: Engine (0 = V-shaped, 1 = straight)

Fitting a Gaussian linear model in R produces the following output

	Estimate	Std. Error	t value	Pr(> t)
β_1 (Intercept)	15.5731	16.3817	0.951	0.3517
β_1 wt	-3.9437	1.2874	-3.063	0.0055
β_2 am = 1	2.7937	1.8682	1.495	0.1484
β_3 cyl	-0.2786	0.9348	-0.298	0.7683
β_4 disp	0.0147	0.0120	1.223	0.2338
β_5 hp	-0.0214	0.0162	??	0.1995
β_7 drat	0.8151	1.5101	0.540	0.5946
β_8 qsec	??	0.6587	1.229	0.2314
β_9 vs = 1	0.3684	2.0116	0.183	??

Residual standard error $\sqrt{\sum_{i=1}^{32}(y_i - \hat{y}_i)^2/23} = 2.544$
Coefficient $R^2 = 0.8678$

- a) Write the statistical model corresponding to the analysis (quantities and assumptions). Denote this model as “model A”.
- b) Write the parameter space and sample space.
- c) Complete the missing values in the table.
For “Pr(>|t|)” of vs1, write the meaning of the missing value and how to obtain it.
- d) Perform a test of overall significance of the model using a 5% significance level.

e) On the same dataset, it is then estimated a reduced model (“model B”) that only includes the variables wt and am. The software estimates the following quantities:
Residual standard error = 3.098
Coefficient $R^2 = 0.7528$
What procedure would you use to compare model A and model B? Following your chosen procedure, which model do you prefer?

f) Starting from model B, it is then introduced as additional covariate the interaction between wt and am. Explain the resulting model and how you interpret the parameters.

d) $\alpha = 0.05$

$$\begin{cases} H_0: \beta_2 = \beta_3 = \beta_4 = \beta_5 = \beta_6 = 0 \\ H_1: \text{1 Ho} \end{cases}$$

$$f_{0.05} \sim f_{p-p_0; n-p} \quad f_{0.05} = \frac{\frac{SSR}{p-p_0}}{\frac{SSE}{n-p}} = \frac{\frac{977}{8}}{\frac{112.85}{23}} = 18.87$$

$$\bullet \sqrt{\frac{1}{23} \sum (y_i - \hat{y}_i)^2} = 2.544 \rightarrow SSE = (2.544)^2 \cdot 23 = 148.85$$

$$\bullet R^2 = \frac{SSR}{SST} = \frac{SST - SSE}{SST} \rightarrow SST \cdot R^2 = SST - SSE \rightarrow SST(R^2 - 1) = -SSE \rightarrow SST = \frac{-SSE}{R^2 - 1} = 112.6$$

$$\bullet SSR = SST - SSE = 977$$

$$\alpha^{obs} = P(F_{p-p_0; n-p} | f_{obs}) \approx 1 - 0.9999 \approx 0 \quad \text{Since } \alpha^{obs} < \alpha \text{ for } \alpha = 0.1; \alpha = 0.05; \alpha = 0.01 \mid \text{REJECT } H_0 \text{ AT ALL CONFIDENCE LEVEL}$$

e) Mod A:

$$Y_i = \beta_1 + \beta_2 x_{i1} + \beta_3 0_{2i} + \beta_4 x_{i3} + \beta_5 x_{i4} + \beta_6 x_{i5} + \beta_7 x_{i6} + \beta_8 x_{i7} + \beta_9 0_{9i}$$

$SSE = 148.85$

$SSR = 977$

$SST = 1126$

Mod B:

$$Y_i = \beta_1 + \beta_2 x_{i1} + \beta_3 0_{2i}$$

$SSE = (\text{RESIDUAL ST. ERROR})^2 \cdot (n-p)$

$$SSE = (3.098)^2 \cdot (n-p_0) = (3.098)^2 \cdot 29 = 278.33$$

$R^2 = 0.7528 \rightarrow SSR = R^2 \cdot SST = 847.6$

$SST = 1126$

NESTED MODEL, USE F TEST

$$\begin{cases} H_0: \beta_4 = \beta_5 = \beta_6 = \beta_7 = \beta_8 = \beta_9 = 0 \\ H_1: \text{1 Ho} \end{cases} \quad f_{0.05} \sim f_{p-p_0; n-p}$$

$$f_{0.05} = \frac{\frac{SSE_A - SSE_B}{p-p_0}}{\frac{SSE_B}{n-p}} = \frac{\frac{278.33 - 148.85}{9-3}}{\frac{148.85}{23}} = 3.33$$

$$\alpha^{obs} = P(F_{p-p_0; n-p} | f_{obs}) \approx 0.018$$

SINCE $\alpha^{obs} < \alpha$ FOR $\alpha = 0.1$ AND $\alpha = 0.05$ I REJECT H_0 AT THOSE CONFIDENCE LEVEL
SINCE $\alpha^{obs} > \alpha$ FOR $\alpha = 0.01$ I DO NOT REJECT H_0 AT THIS CONFIDENCE LEVEL

f) Mod B:

$$Y_i = \beta_1 + \beta_2 x_{i1} + \beta_3 0_{2i} + \beta_{10} x_{i1} : 0_{2i}$$

IF MANUAL
 $\mu_i = \beta_1 + \beta_2 x_{i1} + \beta_3 + \beta_{10} x_{i1}$

IF AUTOMATIC
 $\mu_i = \beta_1 + \beta_2 x_{i1}$

AUTOMATIC

- β_1 INTERCEPT
- IF I INCREASE WEIGHT BY ONE UNIT FOR AUTOMATIC CAR, ON AVERAGE, THE MEAN CONSUMPTION WILL INCREASE BY β_2

HANDAL

- $\beta_1 + \beta_3$ INTERCEPT
- IF I INCREASE WEIGHT BY ONE UNIT FOR MANUAL CAR, ON AVERAGE, THE MEAN CONSUMPTION WILL INCREASE BY $\beta_2 + \beta_{10}$